



Mental Health Habit Tracker

Principles of Software Design Project

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1 Project Description

1.1 Introduction

Mental health is an important part of overall well-being. In daily life, many people experience stress, anxiety, and emotional ups and downs. However, most individuals do not regularly track their emotions or habits, which makes it difficult to understand patterns that affect their mental health. The **Mental Health Habit Tracker** is designed to help users become more aware of their emotional state and daily habits. The application allows users to log their mood, create healthy habits, and track their progress over time. By collecting this information, the system can provide helpful insights about mood trends and habit completion. The system is designed using a **modular approach**, where each module is responsible for a specific function. These modules include the Tracking Module, Analytics Module, Notification Module, and Recommendation Module. This structure makes the system easier to manage, improve, and expand in the future.

1.2 Problem Statement

Many people want to improve their mental well-being and daily routines, but they often struggle to stay consistent. People may forget to record their mood, fail to maintain healthy habits, or simply lack the tools needed to understand their emotional patterns. Traditional methods such as writing in journals or using simple reminder apps do not provide deeper insights into mood changes or habit performance. As a result, users may not recognize which habits positively affect their mental state or when they experience emotional changes. Because of this, there is a need for a system that not only tracks moods and habits but also analyzes this data, sends reminders, and offers helpful suggestions that support better mental health.

1.3 Objectives

The main goal of the Mental Health Habit Tracker is to support users in building healthier habits and becoming more aware of their emotional well-being. The project aims to:

- Allow users to easily **record their daily mood**.
- Help users **create and track positive habits**.
- Store user information in a structured database for future analysis.
- Analyze collected data to identify **patterns in mood and habits**.
- Provide users with **visual insights**, such as charts showing mood trends and habit completion rates.
- Send **reminders and notifications** to encourage users to stay consistent.
- Offer **simple recommendations** that may help improve the user's mental health.
- Use a **modular system design** that keeps the application organized and easier to develop.

1.4 Target Users

The Mental Health Habit Tracker is intended for anyone who wants to better understand their emotions and improve their daily habits. The application can be useful for different types of users, including:

- **Students** who may experience stress from studying and exams.
- **Working professionals** who want to manage stress and maintain a healthy routine.
- **People interested in personal development** who want to track their habits and progress.
- **Anyone focusing on mental wellness** and looking for a simple way to monitor their mood and daily activities.

2 Description of Each Service / Module

2.1 Front-End (User Interface)

The front-end represents the visual and interactive layer of the Mental Health Habit Tracker system. It was designed to provide users with a calming, supportive, and engaging experience that encourages mental wellness, emotional awareness, and habit consistency. The interface focuses on simplicity, accessibility, and emotional comfort while allowing users to easily navigate between different wellness features.

2.1.1 Landing Page

The landing page serves as the introduction to the platform and welcomes new users to the system. It presents the main purpose of the application, which is supporting users in improving their mental well-being through mood tracking, habit building, analytics, and personalized wellness recommendations.

2.1.2 Authentication Pages

The system includes secure authentication interfaces that allow users to create and access personal accounts.

The authentication system includes:

- A Log In page containing:
 - Email
 - Password
 - Forgot password functionality
- A Sign Up page where users register using:
 - Name

- Email
- Password

2.1.3 Dashboard Page

The dashboard acts as the central hub of the application and provides users with an overview of their mental wellness activities and progress.

The dashboard includes:

- A moving orb animation built using React to encourage breathing and relaxation.
- A sanctuary map that visually displays all website sections and redirects users between pages.
- Daily habits progress tracking.
- Daily streak monitoring.
- Daily motivational quotes.
- Micro-actions that redirect users to wellness suggestions.
- Daily logs displaying:
 - Saved moods
 - Completed habits
- A reflections section where users can write and save personal notes.

2.1.4 Mood Page

The Mood Page allows users to record and monitor their emotional state over time.

The page includes:

- Daily mood tracking and saving.
- Mood streak monitoring.
- Mood history and analytics.
- Charts displaying mood statistics:
 - Daily
 - Monthly
 - All-time

Supported moods include:

- Happy
- Low
- Stressed
- Neutral
- Angry

Additional wellness features include:

- Meditation recommendations for:
 - Better sleep
 - Anxiety relief
 - Morning meditation
 - Mindfulness
 - Concentration
- A mood calendar that displays emojis depending on the mood logged each day.

2.1.5 Habits Page

The Habits Page allows users to create, customize, and manage healthy routines using the “Add Habit+” feature.

Users can:

- Create habits.
- Select habit frequency:
 - Daily
 - Weekly
 - Monthly

Depending on the selected frequency:

- Daily habits allow users to specify how many times the habit should be completed per day.
- Weekly habits allow users to select repeat days during the week.

- Monthly habits allow users to define completion goals per month.

Additional customization options include:

- Duration choices:
 - 5 minutes
 - 10 minutes
 - 15 minutes
 - 30 minutes
 - 1 hour
- Reminder scheduling and time customization.

2.1.6 Notifications Interface

The notifications interface displays important user reminders and system updates.

Displayed notifications include:

- Successful sign-up notifications.
- Habit logging notifications.
- Habit completion notifications.
- Reminder notifications.
- “We miss you” notifications for inactive users.

2.1.7 Analytics Page

The Analytics Page visually presents user wellness data and behavioral insights.

The page includes:

- Completion rate tracking.
- Overall consistency analysis.
- Active streak monitoring.
- Focus level evaluation based on activity frequency.
- Mood-over-time charts referred to as “Growth Phase”.
- Habit activity tracking.
- Observations from the application mascot summarizing user performance and emotional progress.

2.1.8 Recommendations Page

The Recommendations Page provides personalized wellness activities and emotional support based on the user's selected mood.

Users can select how they feel today:

- Low
- Neutral
- Stressed
- Happy

The page also includes:

- A “Today's Focus” section encouraging users to avoid unnecessary pressure and focus on manageable goals.
- A weekly wellness plan.
- A reinforcement section containing motivational and supportive messages.

Smart Suggestions

Examples of smart suggestions include:

- Morning Reset Walk with:
 - 5-minute timer
 - Pause button
 - Reset button
- Reflection Checkpoint containing guided questions that generate emotional insights.
- Listen to Nature Sounds:
 - Ocean Waves
 - Rain on Windows
 - Wind Through Trees

Each sound option includes calming visuals and continuous audio playback.

Micro Actions

The system also provides small wellness activities intended to quickly improve the user's emotional state.

Examples include:

- Guided deep breathing exercises with inhale/exhale visual timing.
- WhatsApp integration for messaging someone trusted.
- Fresh air activities with a stopwatch tracker.

Stress Relief Features

When users select the “Stressed” mood option, the system displays:

- Pressure release exercises with step-by-step visual guidance.
- Relax shoulders and jaw exercises.
- Additional support recommendations if stress levels remain high.

2.2 Back-End System

The back-end system manages the application's core logic, database communication, analytics generation, recommendation processing, and notification handling. It ensures secure data management and seamless communication between all modules of the system.

2.2.1 Authentication Management

The authentication system is responsible for:

- User registration.
- Login validation.
- Password recovery functionality.
- Secure account authentication and session management.

2.2.2 Tracking Module

The Tracking Module handles all user-generated wellness data.

Responsibilities include:

- Storing mood entries.

- Storing habit information.
- Tracking habit completion.
- Saving reflection notes.
- Logging daily activities and user interactions.

The collected data is stored in the database for future analysis and personalized recommendations.

2.2.3 Analytics Module

The Analytics Module processes stored user data and generates wellness insights and behavioral statistics.

The module is responsible for:

- Mood trend analysis.
- Habit completion rate calculations.
- Active streak calculations.
- Consistency tracking.
- Focus level evaluation.
- Habit activity statistics generation.

The generated analytics are displayed visually through charts and graphs on the front-end interface.

2.2.4 Recommendation Module

The Recommendation Module generates personalized wellness suggestions based on:

- Mood history.
- Habit completion data.
- User activity patterns.
- Analytics results.

The recommendation system dynamically adapts suggestions according to the user's emotional state and behavioral patterns.

2.2.5 Notification Module

The Notification Module manages automated reminders and engagement notifications.

Its responsibilities include:

- Scheduling reminders.
- Sending habit notifications.
- Delivering sign-up confirmations.
- Triggering inactivity reminders.
- Sending wellness encouragement notifications.

2.2.6 Database System

The database stores all application information and supports communication between all system modules.

Stored data includes:

- User accounts.
- Mood entries.
- Habit information.
- Habit completion logs.
- Notifications.
- Recommendations.
- Reflection notes.
- Analytics history.
- User progress and streak information.

The database enables the system to generate personalized insights, maintain user progress records, and support all core functionalities of the application.

3 Use Case Diagram

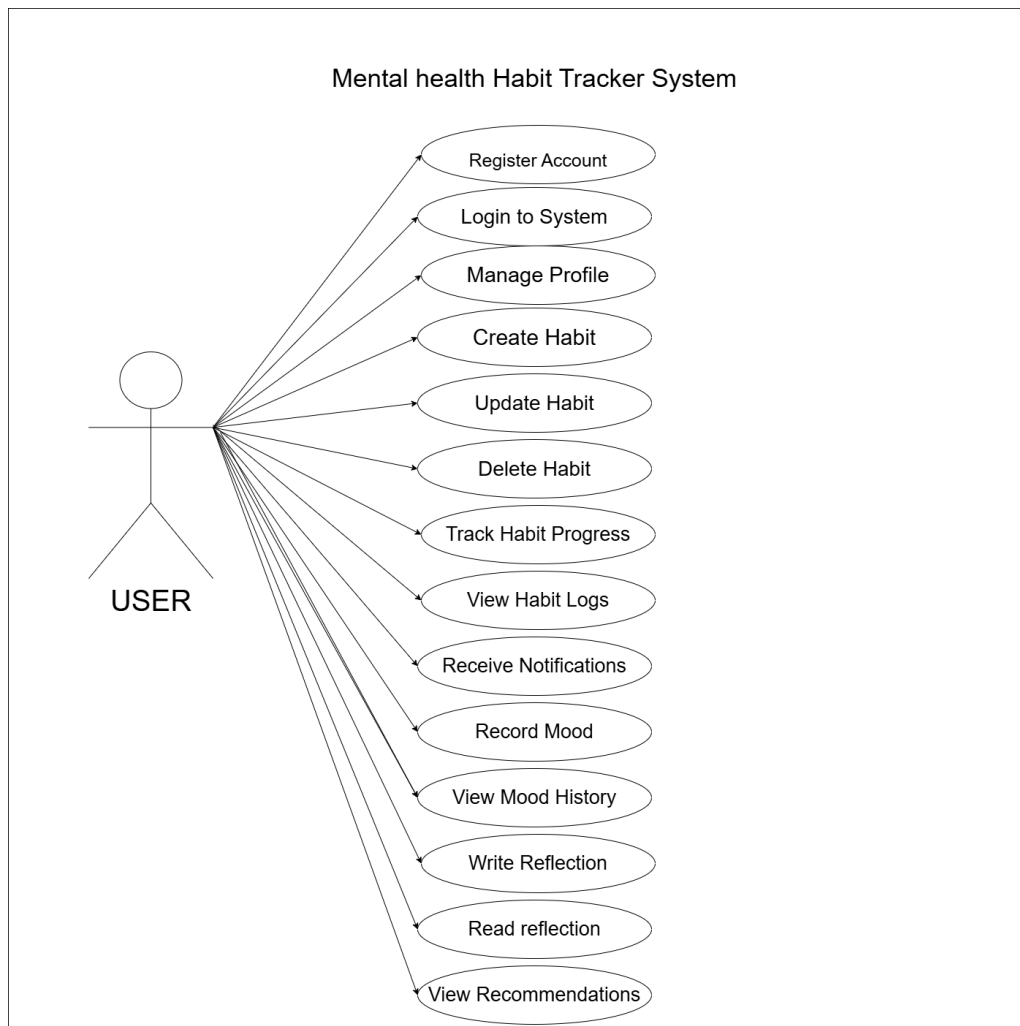


Figure 3.1: Use Case Diagram of the System

4 Entity Relationship Diagram (ERD)

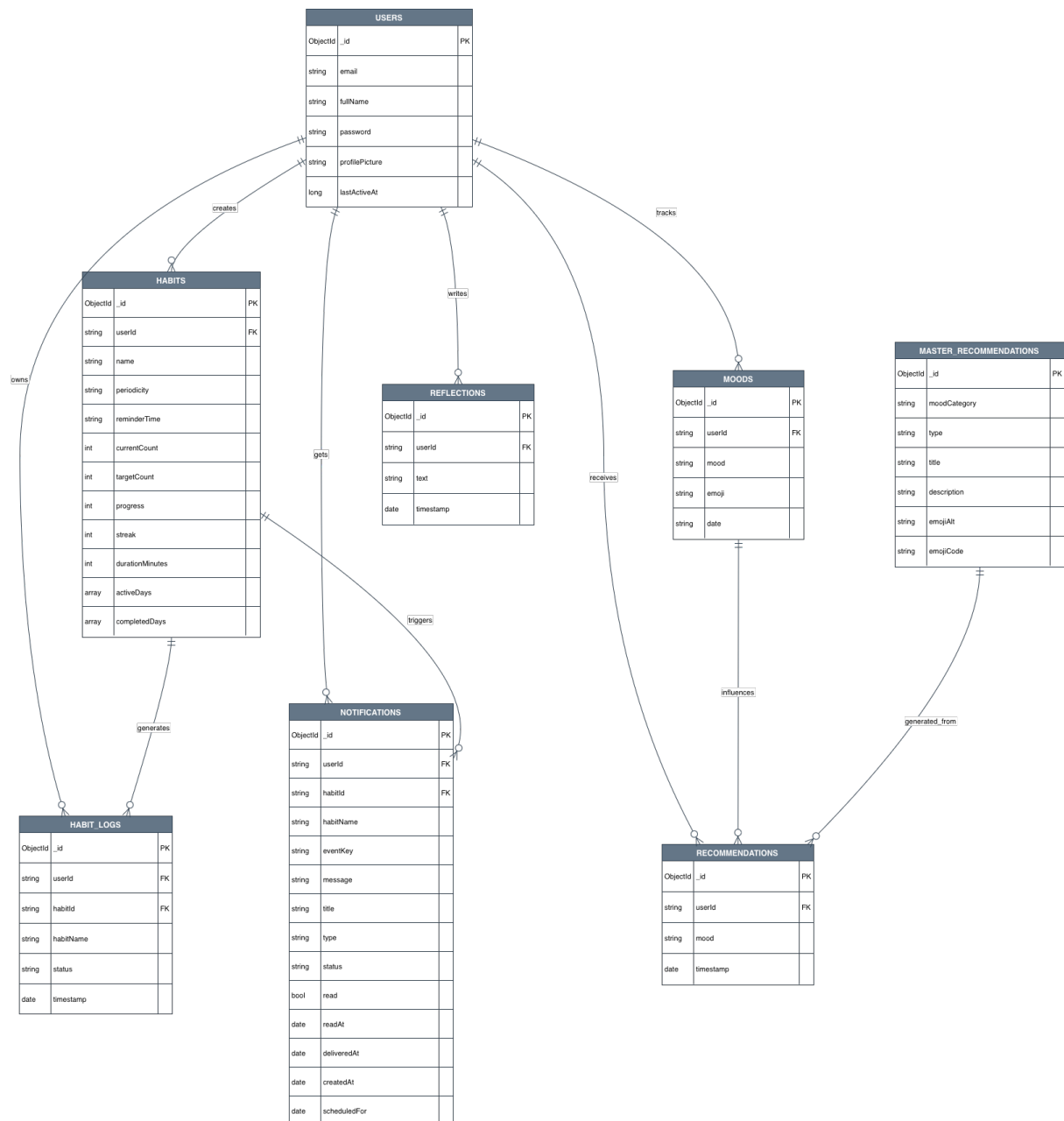


Figure 4.1: Database ERD for the System

5 System Architecture

5.1 System Architecture Diagram

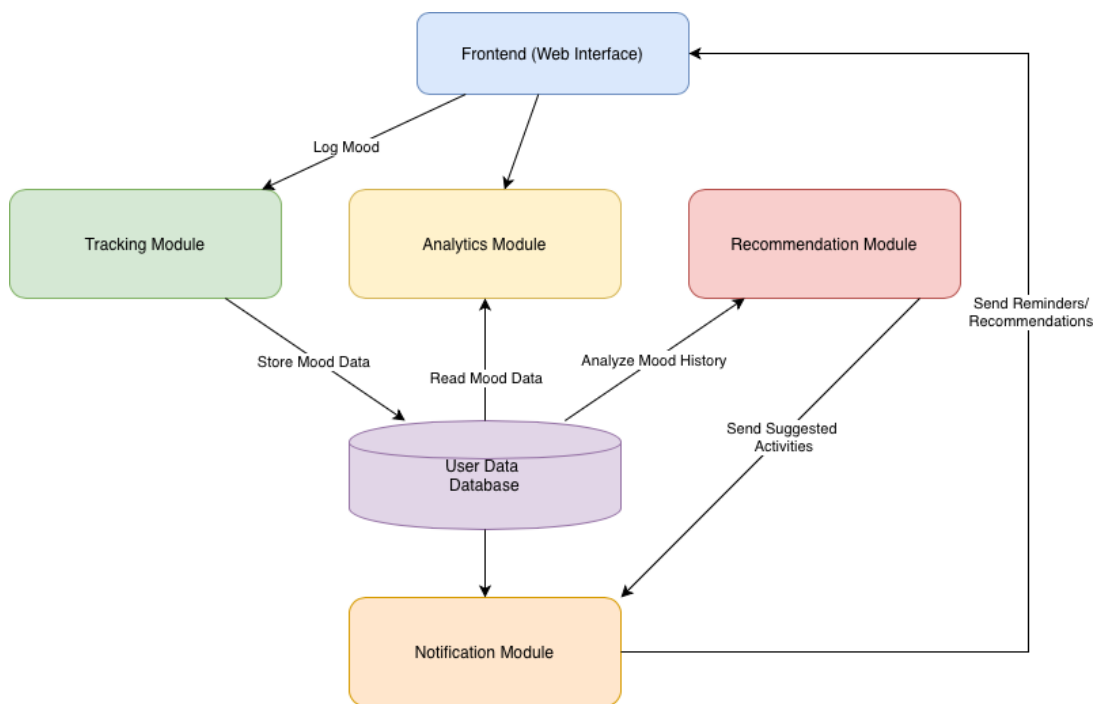


Figure 5.1: System Architecture Overview

The app consists of four main modules: **Tracking**, **Analytics**, **Notification**, and **Recommendation**. The workflow and communication between modules is as follows:

1. **Tracking Module** collects user data
2. **Analytics Module** accesses data from the database
3. **Notification Module** interacts with the database and Analytics Module
4. **Recommendation Module** uses data and analytics to generate recommendations
5. **Workflow Summary:**

User → Tracking Module → Database → Analytics Module →
Recommendations/Notifications → User

5.2 Software Architecture and Design Principles

5.2.1 Design Patterns

Singleton Management

The backend uses Spring Boot's built-in singleton system, which means important classes such as services, controllers, and repositories are created only once and reused throughout the application. This helps save memory and keeps the application organized.

Model-View-Controller (MVC) Architecture

The project follows the MVC structure in both the backend and frontend.

- **Model:** Stores and manages application data such as habits, moods, and users.
- **View:** Displays the user interface using React components.
- **Controller:** Handles user requests and connects the frontend with the backend logic.

Using MVC keeps the project organized and makes the code easier to manage and update.

Builder-Based Configuration

The security settings are created step-by-step using a builder-style configuration. This makes the security code cleaner, easier to read, and simpler to maintain.

Service Layer Abstraction

The project uses a service layer to handle the main business logic. Tasks such as managing habits, sending notifications, and handling emails are processed inside service classes, while controllers focus only on handling requests and responses.

5.2.2 SOLID Principles

Single Responsibility Principle (SRP)

Each class in the project has one specific job. For example, controllers handle requests, services manage business logic, and email classes only send emails.

Open/Closed Principle (OCP)

The system is designed so new features can be added without changing existing code too much. This makes the project easier to improve and maintain.

Liskov Substitution Principle (LSP)

Repository interfaces can be replaced with other implementations without affecting how the application works.

Interface Segregation Principle (ISP)

The project uses smaller and more focused interfaces instead of one large interface with unnecessary methods.

Dependency Inversion Principle (DIP)

The application depends on interfaces and abstractions instead of directly depending on specific implementations. This improves flexibility and makes testing easier.